Integration: Double Angles

Questions 1–16

Question 1.

a) Write $\cos 2x$ in terms of $\cos x$

b) Hence show that $\int_0^{\pi/4} \cos^2 x \, dx = \frac{2+\pi}{8}$

Question 2.

a) Write $\cos 2x$ in terms of $\sin x$

b) Hence show that $\int_0^{\pi/2} \sin^2 x \, dx = \frac{\pi}{4}$

Question 3. Find

a) $\int (1+\sin x)^2\,dx$

b) $\int (1-\cos x)^2\,dx$

c) $\int (\sin x-\cos x)^2\,dx$

d) $\int (\sin x-\cos x)(\sin x+\cos x)\,dx$

Question 4. Find

a) $\int \sin x\cos x\,dx$

b) $\int 8\sin 2x\cos 2x\,dx$

c) $\int 12\sin 5x\cos 5x\,dx$

d) $\int \cos^2 6x\tan 6x\,dx$

Question 5. Evaluate the following definite integrals:

a) $\int_0^{\pi/2}(4-3\sin x)^2\,dx$

b) $\int_0^{\pi/2}(1-6\cos x)^2\,dx$

Question 6. Evaluate the following definite integrals:

a) $\int_0^{\pi/4}(1-8\sin 2x)^2\,dx$

b) $\int_{-\pi}^{\pi}(7+9\cos 3x)^2\,dx$

Question 7. Find

a) $\int (2\cos^2 3x - 1)\, dx$

b) $\int (1 - 2\sin^2 5x)\, dx$

c) $\int (\sin 4x \cot 2x - 1)\, dx$

d) $\int (1 - \sin 20x \tan 10x)\, dx$

Question 8. Find $\displaystyle\int \frac{10\sin x \cos x}{2\cos^2 x + 3}\, dx$

Question 9. The curve y has gradient function

$$\frac{dy}{dx} = \frac{4\tan x}{\sqrt{2\cos 2x + 2}}$$

and passes through the point $(\pi, 6)$.
Show that y can be written in the form $y = a\sec x + b$, where a and b are integers to be found.

Question 10. Evaluate the following definite integral

$$\int_0^{\pi/4} \frac{\sqrt{1-\cos 2x}}{\cos 2x + 1}\, dx$$

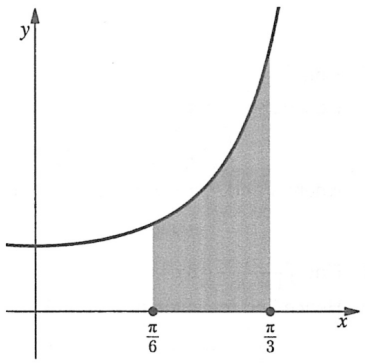
Question 11. Find the exact value of

$$\lim_{\delta x \to 0} \sum_{x=\pi/4}^{\pi/2} \frac{(\cos 2x + 1)\operatorname{cosec} x}{\sin 2x}\,\delta x$$

Question 12. The diagram below shows the curve with equation

$$y = \frac{2\cos x}{\sin 2x(\operatorname{cosec} x - \sin x)}$$

Find the exact value of the shaded area.



Question 13.

a) Use the Compound Angle Formulae to show that $\sin 3x \equiv 3\sin x - 4\sin^3 x$

b) Hence find $\int \sin^3 x \, dx$

Question 14.

a) Use the Compound Angle Formulae to show that $\cos 3x \equiv 4\cos^3 x - 3\cos x$

b) Hence find $\int \cos^3 x \, dx$

Question 15. Prerequisite knowledge for Q15-Q16: Integration by Substitution Evaluate the following definite integral

$$\int_0^{1/2} \sqrt{\frac{16x}{1-x}}\, dx$$

using the substitution $x = \cos^2 u$

Question 16. Evaluate the following definite integral

$$\int_{-1}^{5} \sqrt{(x+1)(5-x)}\, dx$$

using the substitution $x = 2 - 3\cos u$